	GLOBAL ECONOMICS **GRADE: 10** **LEARNING EVIDENCE 2** **DECISION-MAKING UNDER UNCERTAINTY** **TEACHER'S NAME: Nicolás López Cuéllar**	**1ST TERM** **2026–2027**

Learning objective: Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.	**Assessment criteria:** C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss. C6: Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff. C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

Student's name: _____ **Group:** _____ **Date:** _____

1. Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV calculation for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. No points will be awarded if no procedure is shown. A **minimum of 4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	2
8	–	–	2
9	–	–	2
Total	**6**	**6**	**6**

2. 1. Artisan Apiary Expansion [C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits (in thousands of dollars) depend on the honey season.

Alternative	Excellent	Average	Poor
Large	76	34	-14
Medium	58	49	23
Small	38	43	35

Use the Minimax Regret criterion to determine which plan should be selected. Show the best payoff in each state, the regret table, and each alternative's maximum regret.

3. 2. Regional Theater Capacity [C5-2]

A theater operator is choosing a seating-capacity plan. Seasonal profit (in thousands of dollars) is shown below; no reliable probabilities are available.

Alternative	High	Normal	Low
Add two seating sections	108	40	-28
Add one seating section	82	61	14
Lease a nearby hall	59	52	34
Current seating	34	34	34

Determine the decision using Minimax Regret.

4. 3. Emergency Shelter Staffing [C5-2]

A relief organization must select an emergency-shelter staffing plan. Net community value (in thousands of dollars) depends on requests for shelter.

Alternative	Very high	High	Moderate	Low
Full	96	54	24	-24
Flexible	74	64	49	12
Limited	48	51	46	38

Use Minimax Regret to select a staffing plan.

5. 4. Coffee Roastery Product Mix [C6-2]

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

	Strong	Stable	Weak
$P(S_j)$	0.45	0.35	0.20
Single-origin	86	44	-12
Balanced	64	58	22
Classic	42	47	37

Calculate every EMV and apply the Bayes decision criterion.

6. 5. Aquarium Program Scale [C6-2]

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

	High	Expected	Low
$P(S_j)$	0.30	0.40	0.30
Major	98	38	-24
Standard	73	53	18
Compact	52	47	34
Regular tours only	27	27	27

Calculate the EMVs and recommend an alternative.

7. 6. Wind-Farm Maintenance Staffing [C6-2]

A wind-farm operator is choosing a maintenance staffing plan for next quarter. Payoffs are in thousands of dollars.

	Surge	Strong	Steady	Slow
$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring	105	64	24	-32
Moderate hiring	79	68	48	12
Current staff	53	55	51	42

Use EMV to determine the preferred staffing plan.

8. 7. Concert Merchandise Order [C7-2]

A concert vendor must choose between a large merchandise order (A_1) and a small merchandise order (A_2). Payoffs (in thousands of dollars) depend on whether concert attendance is high (S_1) or low (S_2).

Alternative	High (S_1)	Low (S_2)
Large (A_1)	80	20
Small (A_2)	50	40

Let $P(S_1) = p$ and $P(S_2) = 1 - p$. Find the threshold probability and state the preferred alternative for every $0 \le p \le 1$.

9. 8. Meal-Kit Launch [C7-2]

A meal-kit company is comparing a bold launch (A_1) with a measured launch (A_2). Payoffs are in thousands of dollars.

Alternative	High	Moderate	Low
Bold (A_1)	60	30	0
Measured (A_2)	35	45	30

Suppose $P(S_1) = p$ and $P(S_2) = \frac{p}{3}$. Determine the feasible range of p, the threshold, and the preferred launch over the entire range.

10. 9. Publishing-Platform Bid [C7-2]

A publishing firm must choose between a standard platform bid (A_1) and an experimental platform bid (A_2). Payoffs are in thousands of dollars.

Alternative	Immediate	Revision	Delayed
Standard (A_1)	30	70	70
Experimental (A_2)	50	50	30

Suppose $P(S_1) = 3p$ and $P(S_2) = p$. Find the feasible range, threshold probability, and preferred bid for each range of p.